

Super-Eddington Little Blue Dots Reionize Helium Too Early

JWST's faint AGN may be too efficient at ionizing helium, challenging super-Eddington accretion scenarios.

arXiv:2608.28745 — Super-Eddington Little Blue Dots May Reionize Helium Too Early

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Background

JWST found many faint AGN at $z > 4$ with broad lines and blue continua, called Little Blue Dots. Some suggest they are super-Eddington accretors with weak X-rays.

How does super-Eddington accretion in Little Blue Dots affect the timing of helium reionization?

What this paper is selling

The paper shows that if most Little Blue Dots are super-Eddington, helium reionization ends too early, conflicting with observations. Only a small fraction can be super-Eddington.

Key results to remember

- Helium reionization ending after $z=3$ disfavors majority super-Eddington Little Blue Dots.
- Super-Eddington fraction must be $< 10\%$ for fiducial escape fraction and abundance.
- Tighter constraints if bright quasars also contribute to helium reionization.

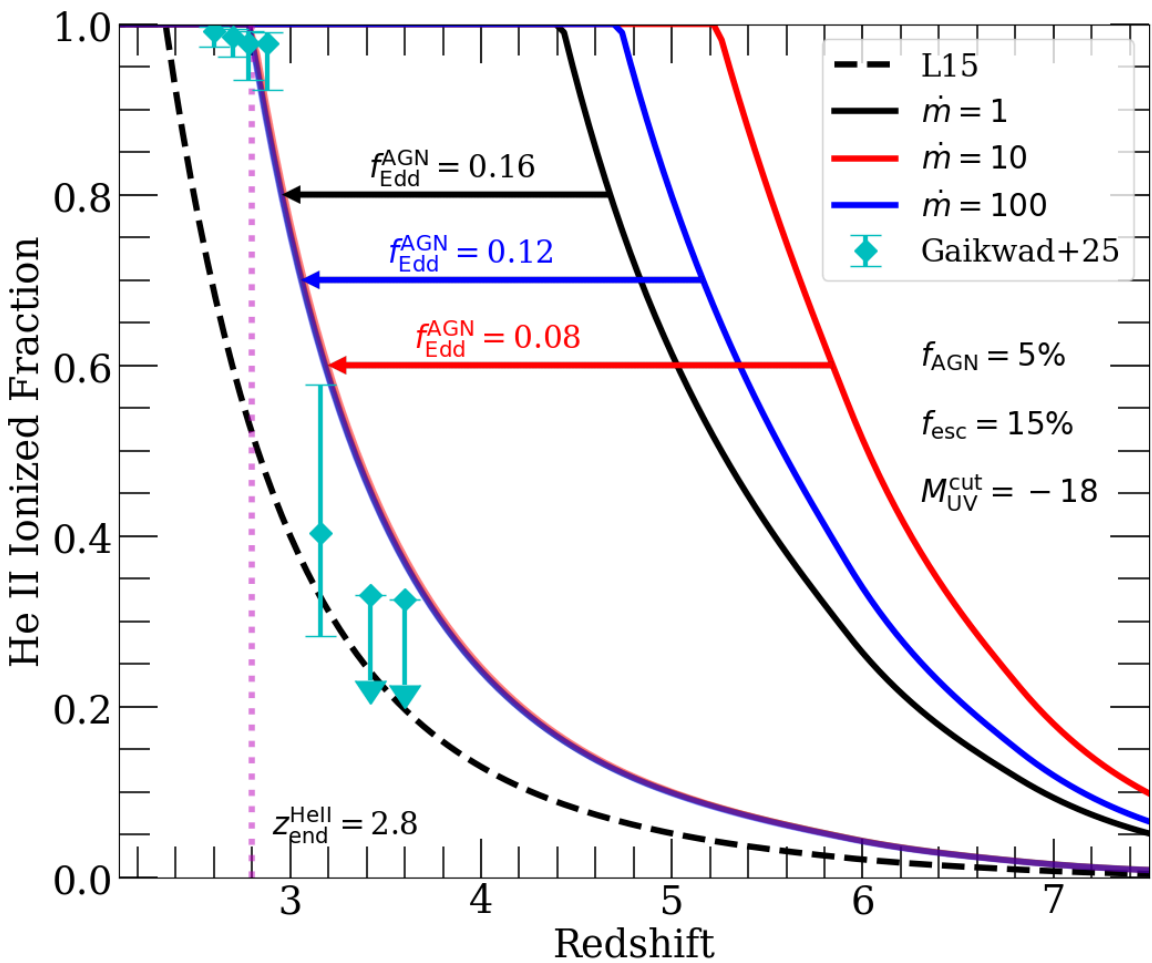


Figure 3 | Helium reionization histories for different SEDs; super-Eddington models complete reionization earlier than observed.

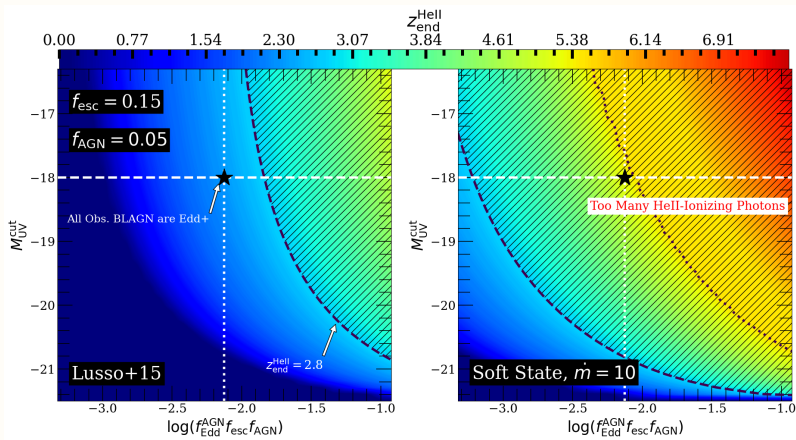


Figure 4 | Excluded AGN parameter space: hatched region shows where helium reionization ends too early.